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Improved Moscovian part of the Gondwana APWP for paleocontinental reconstructions, obtained from a first paleomagnetic pole, age-constrained by a fold test, from In Ezzane area in the Murzuq basin (Algeria, stable Africa)

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ABSTRACT

To improve paleocontinental reconstructions, paleomagnetic reference curves (Apparent Polar Wander Path: APWP) feature for large continents have to be continuously refined by adding up new high-quality data. For stable Africa, the Moscovian period was favorable for such aim, with well-dated and widespread geological formations. A new study has been conducted in the Upper "Dembaba" geological formation of Lower Moscovian age outcropping in the western part of the "Murzuq" basin (Saharan platform). Well-defined ChRMs, combined with remagnetization circles data, both constrained in age by a positive fold test, yield a new significant paleomagnetic pole ($\lambda = 25.2°S$, $\phi = 59.9°E$, $K = 55$, $A_{95} = 5.4°$). When joined with previous African data of the same age, it gives an improved reference pole for Africa ($\lambda = 28.9°S$, $\phi = 54.5°E$, $K = 106$, $A_{95} = 3.6°$). The Mean Moscovian paleomagnetic pole determined from an updated Gondwana Paleozoic APWP ($\lambda = 29.4°S$, $\phi = 51.5°E$, $K = 11$, $A_{95} = 1.8°$), associated with the corresponding Laurussia pole (Domeier et al., 2012), yields a more constrained paleocontinental reconstruction for 310 Ma.

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1. Introduction

To improve the African Apparent Polar Wander Path (APWP), numerous paleomagnetic studies were devoted to the Saharan craton in Algeria during the last decades. Widespread remagnetizations of Permian and Cenozoic ages were pointed out in most works for Lower to Middle Paleozoic ages (Aïfa, 1993; Henry et al., 2004). On the contrary, for the series of Bashkirian age and younger, primary magnetization was established. The obtained results greatly improved the African APWP from the Lower Carboniferous to Liassic times (Derder et al., 2006 and references therein; Kies et al., 1995; Derder et al., 2001d, 2009). However, for paleo-continental reconstructions purpose, precise and reliable comparison of this APWP with that of other continents is necessary, given that present uncertainties on the pole positions remain significant (e.g. McElhinny et al., 2003; Domeier et al., 2012).

The geological formations of well-established Moscovian age outcroping on a large south and eastern parts of the Saharan platform in Algeria (Reggane, Ahnet, Illizi and "Murzuq" basins), constitute the last marine level deposited before a very long period with erosion or deposition of continental facies. For a paleomagnetic purpose, these formations are generally favorable, despite their weak intensity of remanent magnetization (Henry et al., 1992). They represent the single present possibility to obtain a well-defined Upper Paleozoic reference pole in stable North Africa. Different large Saharan basins have already been the subject of detailed paleomagnetic studies for Upper Carboniferous formations, in central Algeria, the Ahnet and Reggane basins (Daly and Irving, 1983; Derder et al., 2001a, 2009) and in eastern Algeria, the Illizi basin (Henry et al., 1992; Derder et al., 2001b, 2001c, 2001d), 800 km apart (Fig. 1a). The last area in Algeria for a new paleomagnetic study of marine Moscovian rocks is located at In Ezzane area, on the western border of the "Murzuq" basin, about 600 Km SSE

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[1] Now at geoexplo: http://www.geoexplo.dz.

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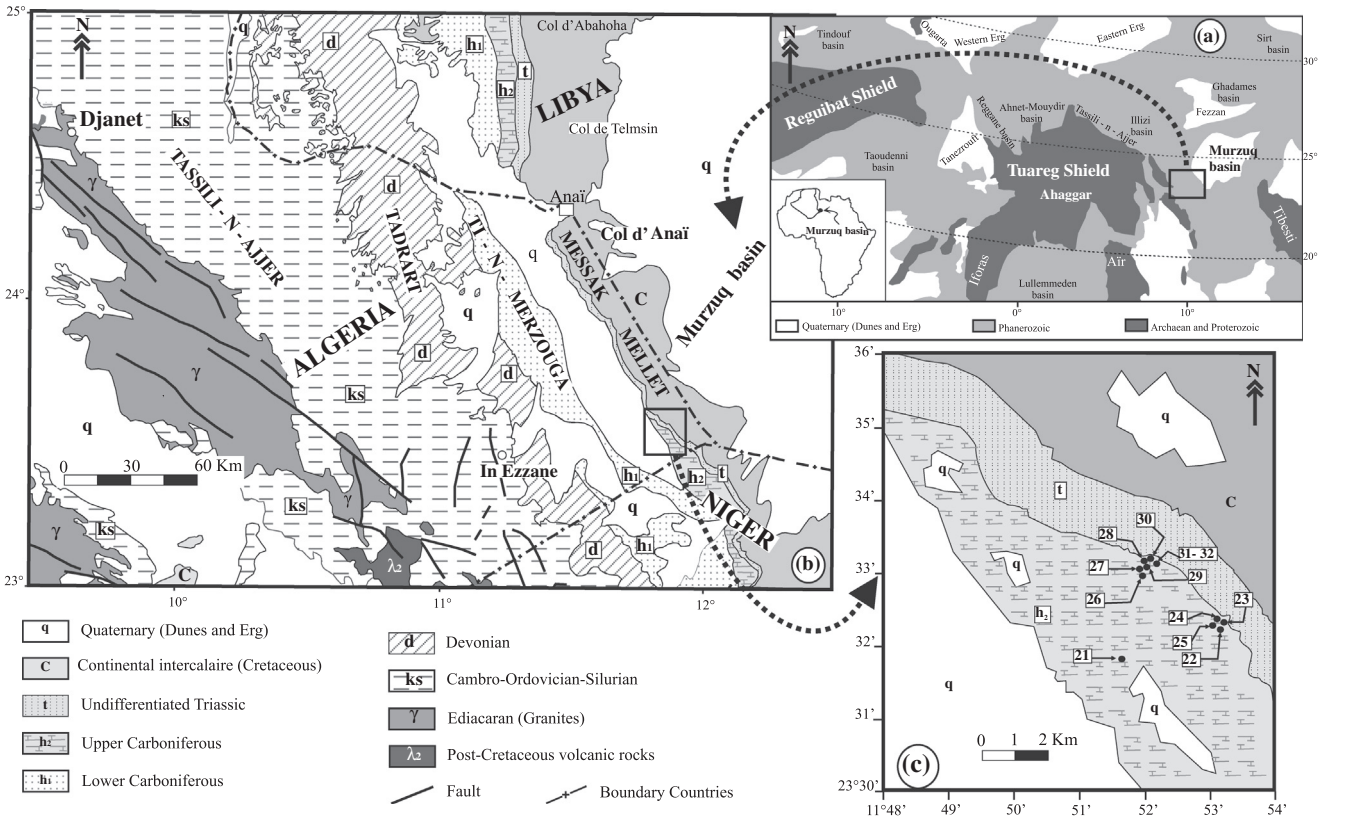


Fig. 1. (a) Location of the studied area in the North Africa context, (b) geological map of the studied area (after "carte géologique du nord ouest de l'Afrique", "carte du Sahara central", 1962; Fezaa et al., 2010), and (c) location of the sampling sites.

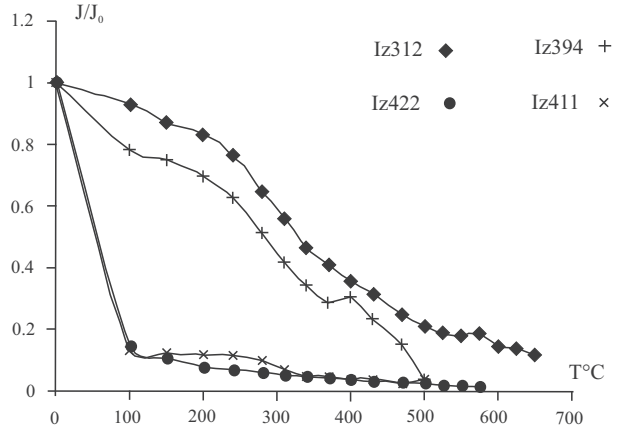


Fig. 2. Thermal demagnetization curves for samples IZ312, IZ394, IZ411 and IZ422.

from the Illizi basin and more than 1000 Km apart from the Ahnet and Reggane basins (Fig. 1a). Though its key position in the Saharan platform, this area was not previously studied because of its very isolated location and difficult access. A fieldtrip for paleomagnetic sampling was then organized on the borders of Algeria, Libya and Niger to obtain a new and complementary datum to a better characterization of the Moscovian pole of the African APWP.

2. Geological setting

The "Murzuq" basin located in central North Africa, in Algeria, Libya and Niger (Fig. 1a) is a key area within the Saharan metacraton (Liégeois et al., 2013). It is one of the largest intracratonic sedimentary basins of the Saharan platform, filled by different Phanerozoic series (see Bellini and Massa, 1980; Fröhlich et al., 2010) covered by a very large sandy desert (the "Murzuq erg"). On its western Algerian border, the Paleozoic marine series are overlain by continental levels, attributed to the "Zarzaïtine" formation of Triassic age (Fig. 1b). The oldest Carboniferous marine formation is the Upper "Dembaba" formation, outcropping in Algeria only in a restricted area at its extreme eastern part. It is mainly constituted by grey and pink fine limestones and marls. It is well-dated of Lower Moscovian age, by a rich fauna (Massa et al., 1988; Mergl and Massa, 1992, 2000; Mergl et al., 2001), in particular brachiopods (*Rugosochonetes* cf. *chesterensis* – Weller – *Enteletes lamarcki*, *Linoproductus cora*, *Concardium rostratum*,...), cephalopods (*Metacoceras*, *Paradomatoceras*) and conodonts (*Ignodiathodus delicates*, *Neognath bassieri*) associated with foraminifers (*Profusinella pseudolibrovichi*, *Aljutovella tikhonovichi*,...).

The different formations in the studied area are tabular with a weak dip toward ENE (about 4° for the studied levels). However, a very local (23°33′04″N, 11°52′16″E) deformation has been observed for the Moscovian formation, with a northwest dip reaching up to 40°. Unfortunately, outcrop conditions did not allow a detailed analysis of this structure and to determine if the overlying continental levels are affected by this folding. A significant regional event then occurred after the deposition of the "Dembaba" formation and the observed local deformation could be a far-effect of the Hercynian orogeny. Similar evolution is well-known in the "Edjeleh" (Illizi basin) area where the equivalent of the "Dembaba" formation ("El Adeb Larache" formation) is strongly folded under the Triassic "Zarzaïtine" formation (see Derder et al., 2001c).

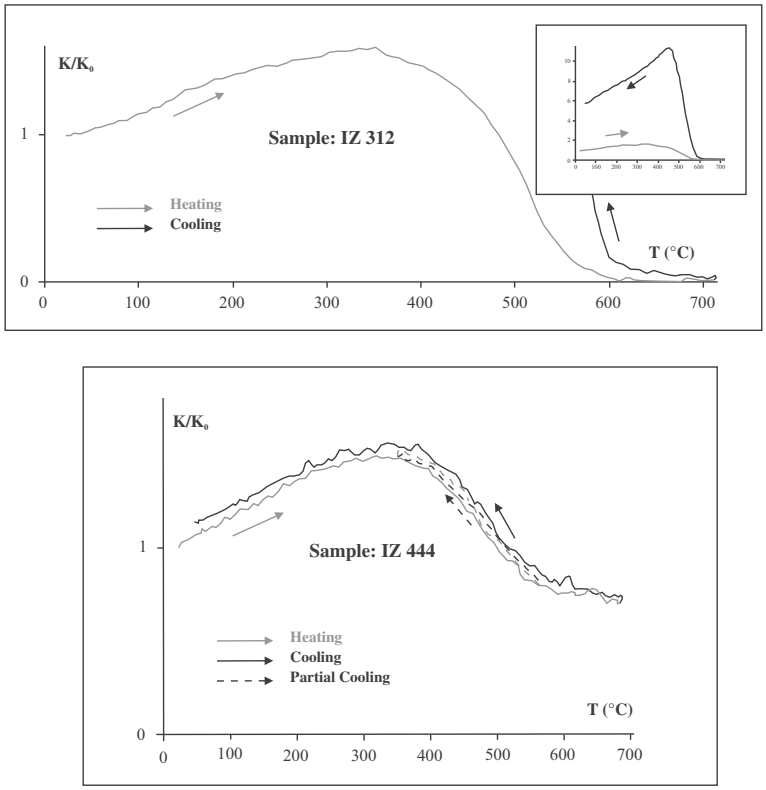


Fig. 3. Typical thermomagnetic curves: normalized low field susceptibility (K/K_o) as a function of the temperature T (in °C) for samples IZ312 and IZ444.

3. Sampling and analysis procedure

The studied area is located on the western border of the "Murzuq" basin (Fig. 1a) approximately 250 km south east of "Djanet" town. The sampling sites were located in the In Ezzane region, south of "Col d'Anai" area (Fig. 1b). 131 standard paleomagnetic samples (Fig. 1c), taken by a portable gasoline-powered drill and distributed on 12 sites, were collected from the grey and pink fine limestones. One–three specimens were cut from each core in order to have additional specimens for pilot studies and rock magnetic analysis. Prior to any demagnetization analysis, the specimens were put in a zero field for at least one month, in order to reduce existing viscous components. Several pilot specimens from each site were subjected to a stepwise alternating field AF (up to 0.1 T) or thermal (up to 670 °C, if the remaining magnetization intensity was still measurable) demagnetizations in order to characterize their magnetic behavior. The results of the pilot studies showed also that the AF procedure did not allow complete demagnetization. Finally, all the 131 samples were thermally demagnetized in this study, 113 of them giving reliable results. In order to correctly isolate and identify the components of magnetization, numerous steps were performed with increments ranging from 100 °C at the lowest temperatures to 10 °C at the highest temperatures. Demagnetization data analysis was carried out using classical methods: the directions of the magnetization components were plotted on orthogonal vector plots (Wilson and Everitt, 1963; Zijderveld, 1967) and the remaining vectors and vectorial differences of the magnetization on equal-area projections; the remagnetization circles methods (Halls, 1976, 1978; McFadden and McElhinny, 1988) were also used. The direction of the different components was computed using principal component analysis (Kirschvink, 1980). Fisher (1953) statistics was used to determine the mean directions and for the progressive unfolding test.

4. Rock magnetism

Representative samples were chosen to determine their magnetic mineralogy from different magnetic approaches, based on thermal demagnetization curves, thermomagnetic K(T) curves (susceptibility as a function of temperature) and hysteresis loops. Remanent magnetization was measured on AGICO JR5 magnetometer, mean susceptibility and K(T) curves on AGICO KLY3, CSL and CS3 equipments and hysteresis loops on a laboratory-made translation inductometer for small samples (about 3 cm^3) within an electro-magnet.

The mean susceptibility km and Natural Remanent Magnetization (NRM) intensity values are weak in all the studied samples, around 2.7×10^{-6} SI and 1.3×10^{-6} A/m, respectively. During thermal demagnetization at high temperatures, in numerous samples, magnetization intensity becomes very low, explaining why demagnetization analysis was stopped at 500 °C to 580 °C in these cases. Half of samples show a sharp decrease of the magnetization intensity at 100 °C heating (more than 80% of the NRM intensity is removed – Fig. 2); these lower unblocking temperatures suggest goethite as one of the magnetic carriers. At higher temperatures, the decrease is progressive, up to approximately 500–580 °C (Fig. 2 – samples IZ411and IZ422), pointing out magnetite as the probable magnetic carrier (beside possible presence of goethite). The second half of samples shows a more progressive decrease up to either 500 °C or 650–670 °C (Fig. 2 – samples IZ394 and IZ312) arguing for significant hematite content at least in part of the latter samples.

The K(T) curves (Fig. 3) have all the same variation during heating, with a regular increase up to 350–400 °C followed by a decrease, for the greater part up to 500–550 °C (suggesting the presence of magnetite) but with a weaker progressive decrease up to 670 °C (indicating a hematite content). Partial cooling loops

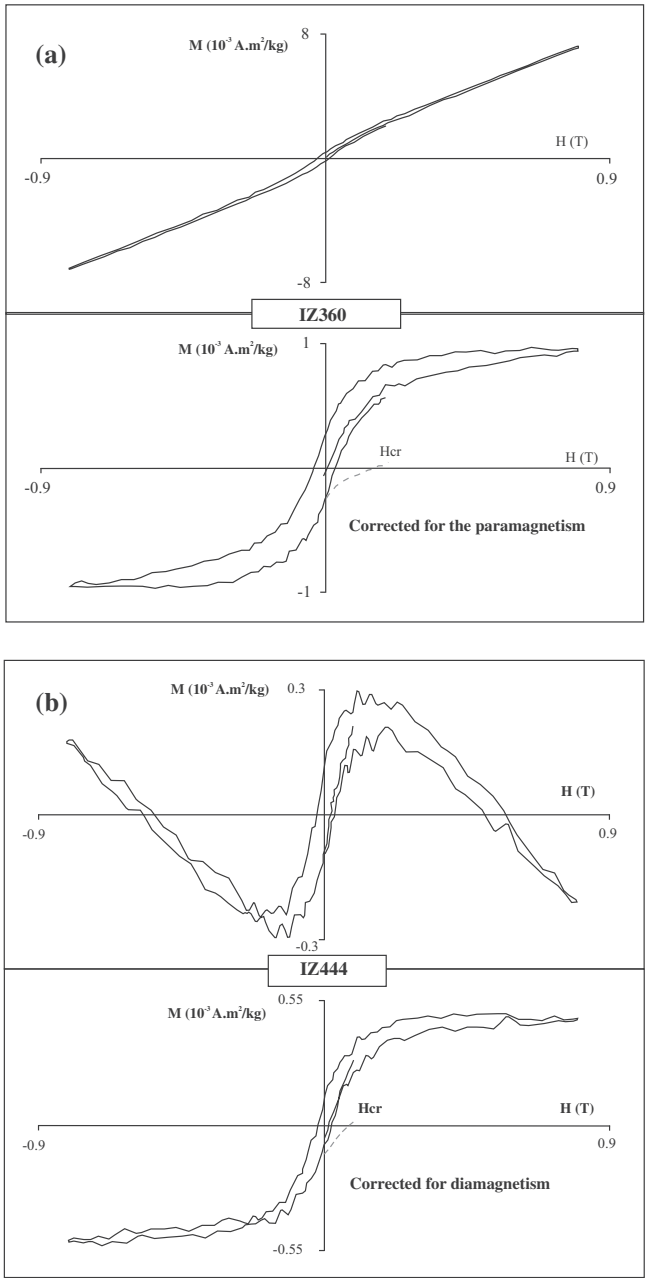


Fig. 4. Typical hysteresis loops after correction (a) for paramagnetism (sample IZ360) and (b) for diamagnetism (sample IZ444). Hcr is the remanent coercive force, Field H in Tesla.

after the heating for temperatures lower than 600 °C show reversible curves. No mineralogical alterations therefore occurred for these samples during heating at these temperatures. At higher temperatures that is also the case for some samples that continue showing reversible curves (Fig. 3 – sample IZ444), while others become irreversible (Fig. 3 – samples IZ312), suggesting the formation of new magnetic phases at very high temperatures. No significant differences were found in curves made under argon or air atmospheres.

Hysteresis loops are mostly not very well-defined because of the weakness of the magnetization. They show a slight wasp waist and are mainly related to paramagnetic (Fig. 4a – IZ360) or diamagnetic (Fig. 4b – IZ444) minerals. They also indicate a partial saturation around 0.1 T (suggesting the presence of a low coercivity component), but do not reach total saturation (presence of a high coercivity component). Remanent coercive force is generally not high (between 0.08 and 0.17 T) confirming a mixing of low and high coercivity components.

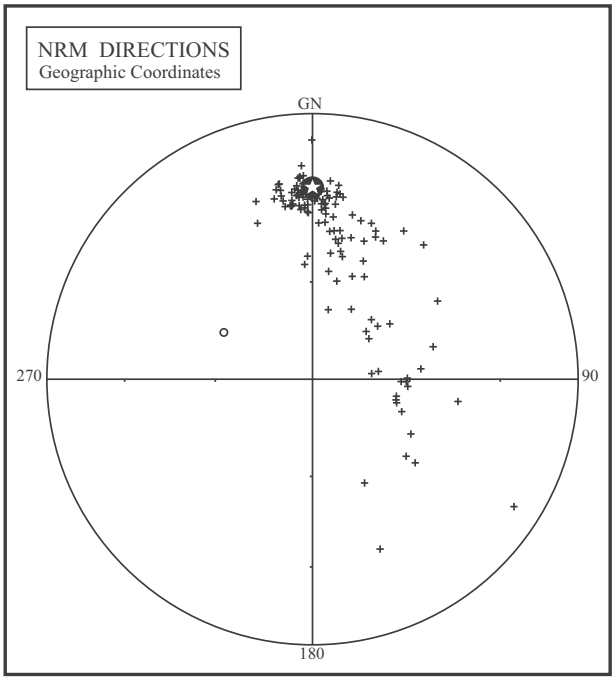


Fig. 5. Equal-area plot of the NRM directions for all samples (crosses: positive inclinations; open circles: negative inclinations). Open star: present magnetic field direction.

The different rock-magnetism experiments therefore suggest that magnetic mineralogy is constituted by magnetite and hematite, besides probably goethite.

5. Paleomagnetic results

The NRM directions are distributed on a girdle on an equal area plot (Fig. 5). This girdle is elongated between the present Earth magnetic dipole field direction and a SE direction. This display suggests a superimposition of at least two components in the NRM.

During thermal demagnetization, after elimination of a viscous component A, the analysis of the magnetization behavior gives three different kinds of evolution. The first one, which has been obtained for most samples, is illustrated by a stable magnetic direction characteristic of a single component B (Fig. 6a). The second type shows magnetic directions evolving along a great circle toward the SE quadrant and reaching a "stable end point" direction B for the highest temperatures (Fig. 6b). Finally, the third one is characterized by directions evolving along a great circle toward the SE quadrant also (superimposition of unblocking temperature spectra for at least two components), but no stable component was reached even for the highest temperatures (Fig. 6c).

The magnetization component B obtained from samples with the two first behaviors is interpreted as a Characteristic Remanent Magnetization (ChRM). It was isolated, between 200 and 400 °C for the threshold of the lowest temperatures and 550–670 °C for the highest ones, for the first case (Fig. 6a), and between 350–400 °C and 500–575 °C for the second one (Fig. 6b). The mean direction of this ChRM B computed from 74 specimens has the following directions: (11 sites): $D = 134.8°$, $I = 36.6°$, $k = 20$, $\alpha_{95} = 3.8°$ and $D = 132.4°$, $I = 38.9°$, $k = 31$, $\alpha_{95} = 3.0°$, before and after dip correction, respectively (Table 1 – Fig. 7). This result is similar to that obtained from the sites mean direction (11 sites): $D = 136.3°$,

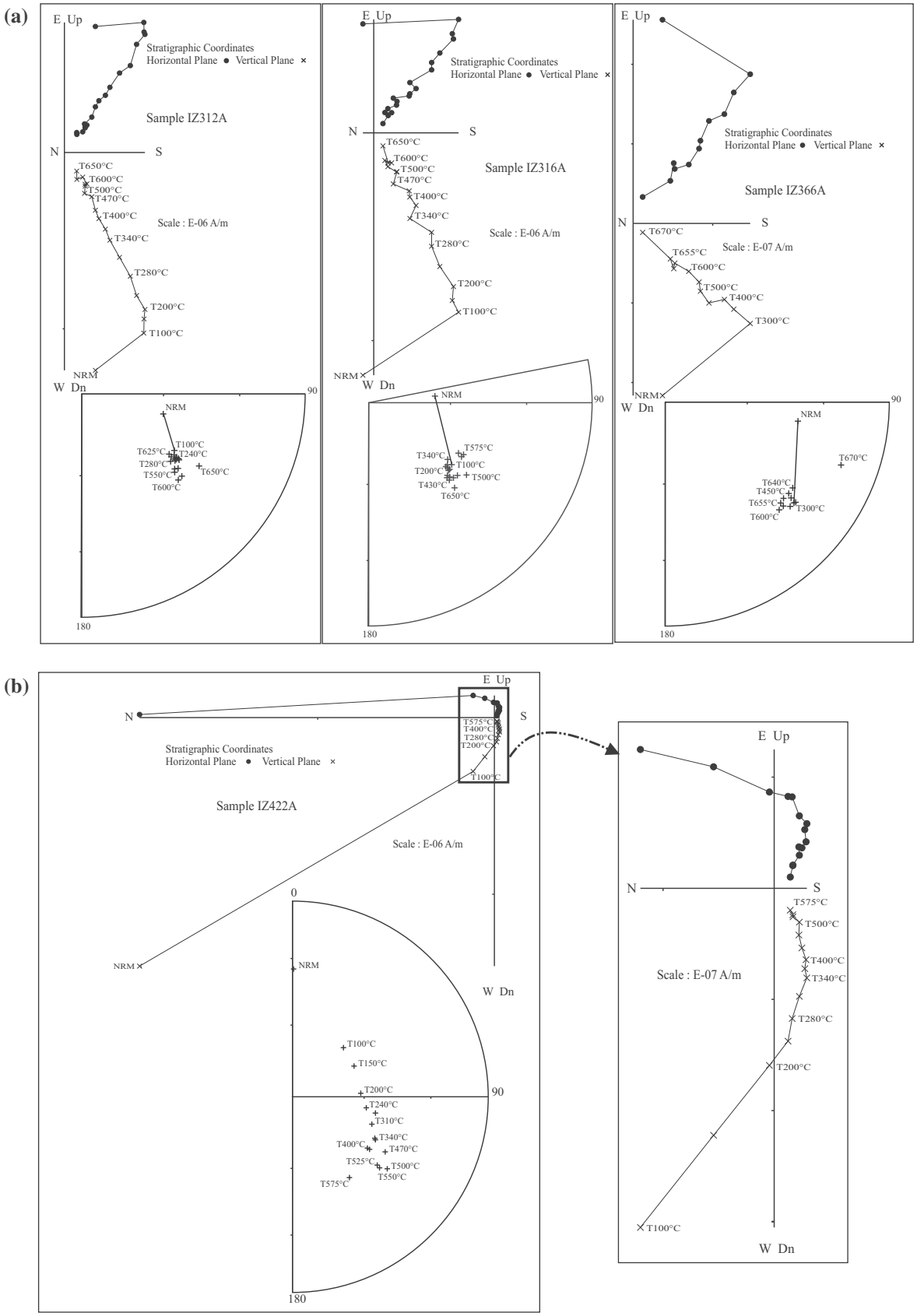


Fig. 6. Orthogonal vector plots (filled circles: horizontal plane, crosses: vertical plane), in stratigraphic coordinates, and their corresponding equal-area plots (crosses: positive inclinations) for (a) samples IZ312A, IZ316A and IZ366A; (b) sample: IZ422A; (c) sample IZ423A.

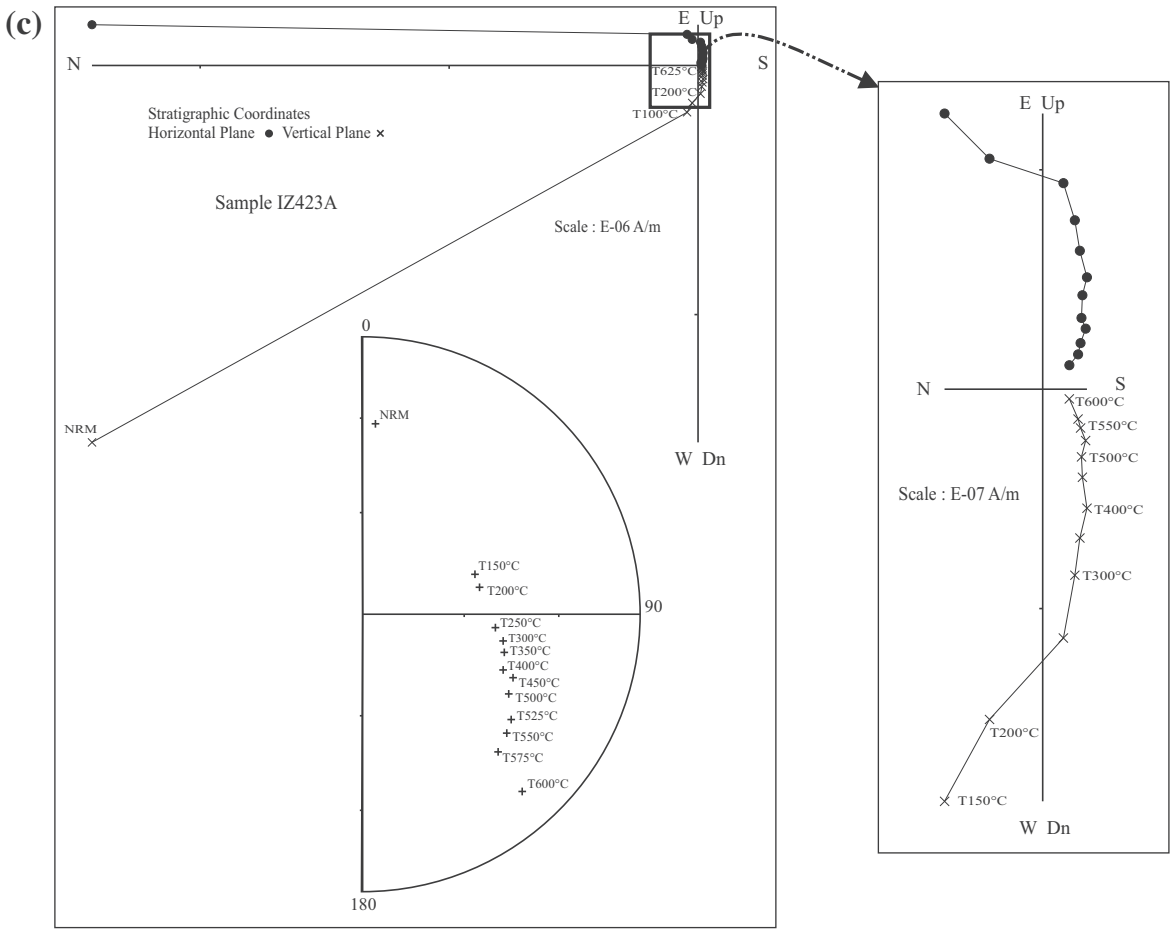


Fig. 6 (*continued*)

$I = 35.4°$, $k = 22$, $\alpha_{95} = 9.9°$ and $D = 133.9°$, $I = 39.3°$, $k = 46$, $\alpha_{95} = 6.8°$. The maximum value of the k precision parameter during progressive unfolding is obtained for 79% unfolding. The difference between k values for 79% (50) and 100% (46) unfolding is statistically not significant, and the simple and stringent McElhinny (1964) fold test is positive and significant at the 95% level.

For samples showing only an evolution along great circles of their magnetization direction without stable end point, the analysis of remagnetization circles using Halls' method (Halls 1976, 1978) was applied. The intersection direction obtained using 39 data distributed in 6 sites is $D = 111.3°$, $I = 52.0°$ and $D = 124.8°$, $I = 43.8°$ respectively before and after dip correction (Table 1). The density diagram of the intersection before bedding correction shows a very large confidence zone at 95% with two maxima at 63%, associated with the recent field direction and a SE dipping direction (Fig. 8). After bedding correction both 95% and 63% zones are much smaller and only correspond to the SE direction. That represents a clear positive fold test (Henry et al., 1999). There is no statistically significant difference between this SE direction obtained from circles intersections and that determined by the ChRM one in stratigraphic coordinates (Table 1). In the single site (27) without any ChRM, the 11 obtained great circles (Table 1) yield a well-defined intersection direction coherent with that of the ChRMs B (Table 1).

These different results show that the McFadden and McElhinny (1988) approach is therefore usable here to integrate great circles and ChRM data within each site. Using this combination of the ChRM B and the great circles data, the mean directions obtained for the different sites are well-clustered (Table 1). The mean direction obtained from all ChRM data ($N = 74$) and all great circles data ($n = 39$) is $D = 133.7°$, $I = 36.1°$, $k = 19.6$ and $\alpha_{95} = 3.1°$ and $D = 131.9°$, $I = 38.8°$, $k = 33.7$, $\alpha_{95} = 2.3°$ before and after dip correction, respectively. This result is very similar to that obtained from the mean sites direction (12 sites – Table 1 – Fig. 9): $D = 135.7°$, $I = 36.0°$, $k = 25.5$ and $\alpha_{95} = 8.0°$ and $D = 133.3°$, $I = 39.3°$, $k = 52.2$, $\alpha_{95} = 5.6°$ before and after dip correction, respectively. During progressive unfolding (Fig. 10) the maximum value of k (57) is obtained for unfolding of 79%. This value is not statistically different from that obtained at 100% unfolding (52). The fold test for these data is again positive and significant (at the 95% level – McElhinny, 1964). The associated paleomagnetic pole is located at $\lambda = 25.2°S$, $\phi = 59.9°E$, K = 55.4, $A_{95} = 5.4°$.

6. Discussion

6.1. A new reference pole for stable Africa

The positive fold test confirms the early character of the B magnetization component isolated in this study, although no proper age determination of deformation affecting the rocks studied is available. The associated pole is very close to a Moscovian pole ($\lambda = 28.3°S$, $\phi = 58.9°E$, K = 157, $A_{95} = 4.2°$ – Derder et al., 2001c) obtained in the Illizi basin at Edjeleh. The paleomagnetic data from Edjeleh also yielded a positive fold test, with outcrop conditions allowing in this area to determine their structural evolution: The folding episode is older than the Upper Triassic, but started before or during the Stephano–Autunian (Attar et al., 1981), i.e. shortly

Table 1
Coordinates of the studied sites (Lat: latitude; Long: longitude in degrees); *P* and *G* (in degrees) are respectively the dip and the azimuth of the down dip direction of the strata (bedding corrections); Mean direction (*N*: number of ChRM B, *n*: number of great circles, Declination *D*; Inclination *I* and corresponding Fisher's parameters *k* and α_{95}) obtained from each site and VGP or paleomagnetic pole (after 100% bedding correction): Lat. (°S), Long (°E), with for the latter corresponding Fisher parameters K and A_{95}. Data for ChRM B (normal font), great circles (italic font) and combined ChRM B and great circles (bold font).

Site	Coord. site		*P*(°)	*G*(°)	Before dip correction					After dip correction				VGPs and Paleomagnetic pole			
					N	*n*	*D*(°)	*I*(°)	*k*	α_{95}(°)	*D*(°)	*I*(°)	*k*	α_{95}(°)	Lat(°S)	Long(°E)	K	A_{95}(°)
	Lat (°N)	Long (°E)															
21	*23°31′46.6″*	*11°51′36.7″*	*4*	*90*	9	–	132.0	53.9	169	4.0	128.7	50.8	169	4.0	16.3	55.7		
22	*23°32′19.2″*	*11°53′15.9″*	*4*	*90*	11	–	133.8	29.0	39	7.4	132.4	26.0	39	7.4	30.4	68.1		
23	*23°32′23.9″*	*11°53′17.6″*	*4*	*90*	8	–	129.7	39.2	33	8.6	127.8	36.0	33	8.6	23.1	65.7		
24	*23°32′26.1″*	*11°53′16.1″*	*4*	*90*	1	–	143.4	39.1	–	–	141.0	36.5	–	–	32.0	56.0		
					–	*11*	*129.3*	*37.9*	–	–	*127.5*	*34.6*	–	–				
					1	**11**	**131.8**	**35.9**	**116**	**4.3**	**130.1**	**32.8**	**116**	**4.3**	**26.1**	**66.0**		
25	*23°32′26.9″*	*11°53′14.4″*	*4*	*90*	7	–	140.0	31.3	69	7.3	138.2	28.6	69	7.3	33.7	62.5		
26	*23°33′02.0″*	*11°52′10.8″*	*4*	*90*	5	–	142.4	30.6	57	10.2	140.7	28.0	57	10.2	35.7	60.8		
					–	2	*0.4*	*29.8*			*2.6*	*29.8*	–	–				
					5	**2**	**145.0**	**31.5**	**40**	**9.9**	**143.2**	**29.1**	**40**	**9.9**	**36.9**	**58.0**		
27	*23°33′04.7″*	*11°52′10.4″*	*4*	*90*	–	*11*	*136.1*	*43.7*	–	–	*133.3*	*41.2*	–	–	*24.6*	*59.0*		
28	*23°33′04.8″*	*11°52′11.5″*	*4*	*90*	10	–	142.2	45.0	168	3.7	139.3	42.4	168	3.7	27.8	54.0		
29	*23°33′04.7″*	*11°52′13.2″*	*4*	*90*	*5*	–	156.4	48.6	195	5.5	152.5	46.5	195	5.5	32.2	40.7		
					–	*3*	*157.5*	*44.9*	–	–	*154.0*	*43.1*	–	–				
					5	**3**	**155.9**	**47.6**	**179**	**4.3**	**152.1**	**45.7**	**179**	**4.3**	**32.6**	**41.5**		
30	*23°33′05.8″*	*11°52′14.2″*	*7*	*90*	7	–	122.2	49.8	150	4.9	118.6	43.7	150	4.9	12.9	66.2		
					–	*4*	*158.6*	*1.6*	–	–	*–21.5*	*1.1*	–	–				
					7	**4**	**122.5**	**49.9**	**148**	**3.8**	**118.9**	**43.7**	**150**	**3.8**	**13.2**	**66.1**		
31	*23°33′04.6″*	*11°52′16.1″*	*32*	*320*	7	–	120.5	10.0	85	6.6	114.4	39.5	85	6.6	11.4	71.1		
					–	*8*	*15.4*	*40.6*	–	–	*1.8*	*18.4*	–	–				
					7	**8**	**122.2**	**11.7**	**67**	**4.8**	**116.0**	**41.4**	**67**	**4.8**	**11.9**	**69.1**		
32	*23°33′04.6″*	*11°52′16.1″*	*41*	*320*	4	–	139.5	8.1	42	14.4	138.5	49.1	42	14.4	23.3	50.5		
Mean					74		134.8	36.6	20	3.8	132.4	38.9	31	3.0	–	–	–	–
						39	*111.3*	*52.0*	–	–	*124.8*	*43.8*	–	–	–	–	–	–
					74	**39**	**133.7**	**36.1**	**20**	**3.1**	**131.9**	**38.8**	**34**	**2.3**	–	–	–	–
Mean					12 Sites		**135.7**	**36.0**	**26**	**8.0**	**133.3**	**39.3**	**52**	**5.6**				
															25.2	**59.9**	**55**	**5.4**

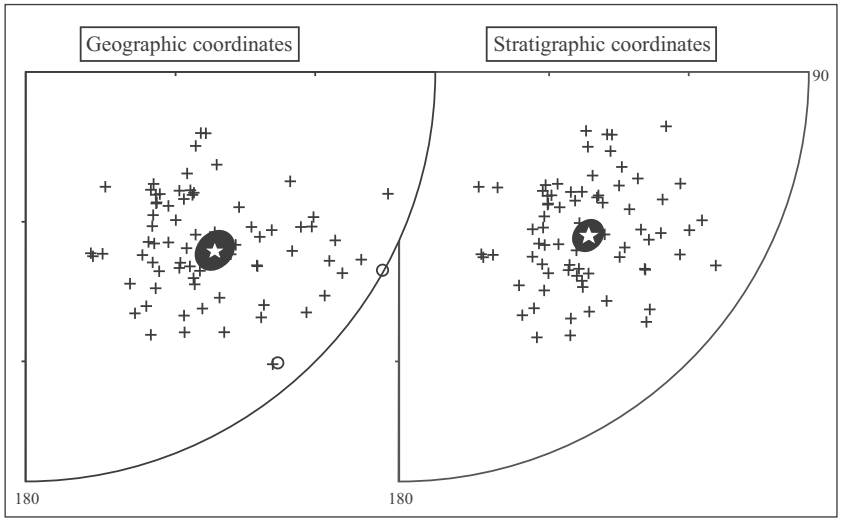


Fig. 7. Equal-area plot of ChRM B, before and after dip correction (crosses: positive inclinations).

after the Moscovian. By analogy, we assume that our Component B (In Ezzane area) represents the primary magnetization and therefore constitutes a new and significant contribution for the Upper Carboniferous paleomagnetic database.

Our new pole is in good agreement with the corresponding segment of the African APWP (Derder et al., 2006). Indeed, it is very close to the paleomagnetic poles for stable Africa from the same and neighboring periods (Fig. 11 and Table 2). The Mean Upper Carboniferous paleomagnetic pole determined from combination of the new pole with the previous ones $\lambda = 28.9°S$, $\phi = 54.5°E$, K = 106, $A_{95} = 3.6°$) can be considered as the Moscovian reference one for Africa.

6.2. Comparison with Gondwana data

The available paleomagnetic data (see list of poles in Derder et al., 2006; McElhinny et al., 2003; Domeier et al., 2012, and Torsvik et al., 2012; and Table 3 as supplementary material in

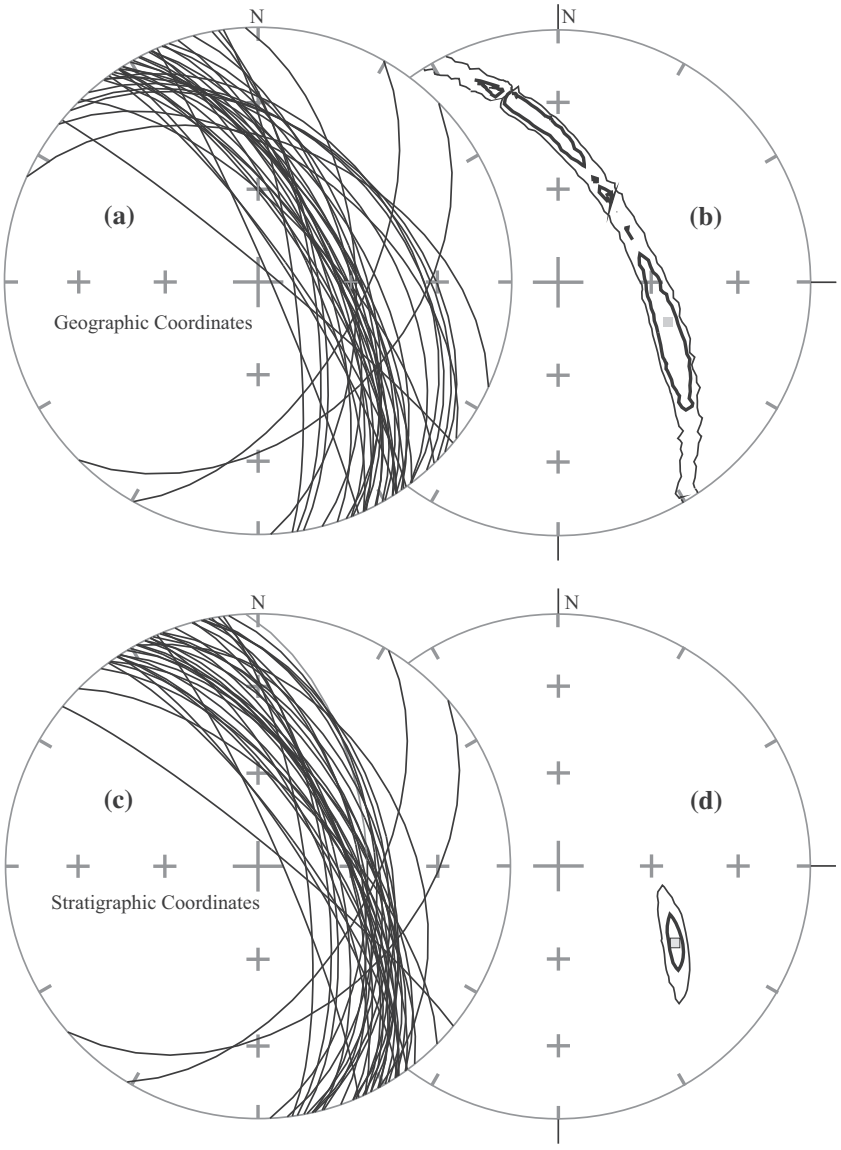


Fig. 8. Remagnetization circles (a, c) and their corresponding confidence zones (Henry et al., 1999) at 63% (thick line) and 95% (thin line) for their best intersection (b, d) obtained from 39 specimens distributed over 6 sites (equal-area plot, lower hemisphere), in geographic (a, b) and stratigraphic (c, d) coordinates.

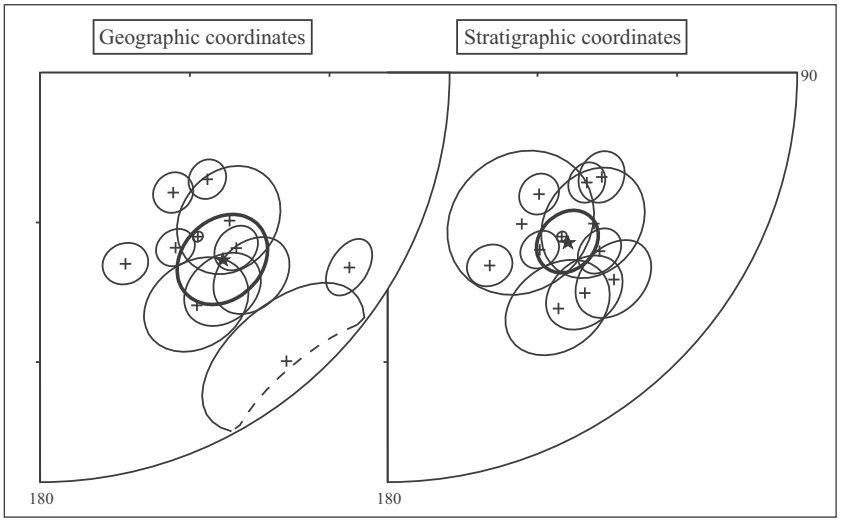


Fig. 9. Equal-area plot of the mean-site paleomagnetic directions with associated 95% confidence zone, in geographic and stratigraphic coordinates (see Table 1).

Table 2
Upper Carboniferous paleomagnetic poles for stable Africa, No refers to number pole in Fig. 11.

Age	No.	Location	λ (°S)	ϕ (°E)	K	A_{95}	References
Stephanian	1	Merkala, Algeria	32.4	56.6	399	2.3	Henry et al. (1999)
Lower/Upper carboniferous?	2	Dwyka Varves, (Tanzania, Zambia)	26.9	30.3	73	10.8	McElhinny and Opdyke (1968)
Early Carboniferous – Early Permian	3	Dwyka group, (Zimbabwe, Tanzania)	27.3	64.8	6.2	12.0	Opdyke et al. (2001)
Late Carboniferous – Early Permian	4	Sakao, Madagascar	33.4	50.0	36.2	9.5	Rakotosolofo et al. (1999)
Upper Carboniferous	5	Sinai, Egypte	31.2	60.8	43.2	7.1	Abdeldayem et al. (1994)
Moscovian	6	Edjeleh, Algeria	28.3	58.9	157	4.2	Derder et al. (2001c)
Moscovian	**7**	**In Ezzane, Algeria**	**25.2**	**59.9**	**55.4**	**5.4**	**This Study**
Moscovian	8	El Abed Larache, Algeria	28.7	55.8	235	2.9	Henry et al. (1992)
Upper Namurian – Lower Moscovian	9	Hassi Bachir, Algeria	32.8	55.7	328	2.0	Derder et al. (2009)
Serpukhovian – Moscovian	10	Ain Ech Chebbi, Algeria	26.5	44.7	383	4.7	Derder et al. (2001a)
Bashkirian	11	El Abed Larache, Algeria	28.2	55.5	207	3.4	Derder et al. (2001b)
Upper Namurian – Lower Moscovian	12	Ain Ech Chebbi, Hassi Bachir, Algeria	25.4	54.8	149	3.5	Daly and Irving (1983)
Westphalian – Namurian	13	Saida, Béchar, Algeria	26.7	57.9	71	9.1	Abou Deeb and Tarling (1984)
Namurian	14	Rouina, Algeria	28.4	56.9	642	1.7	Merabet et al. (1999)

The results of the present study are in bold.

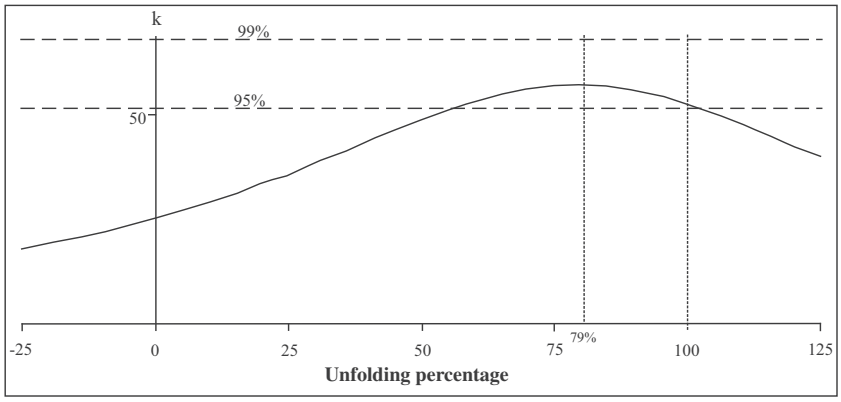


Fig. 10. Variation of *k* (Fisher, 1953) parameter during progressive unfolding.

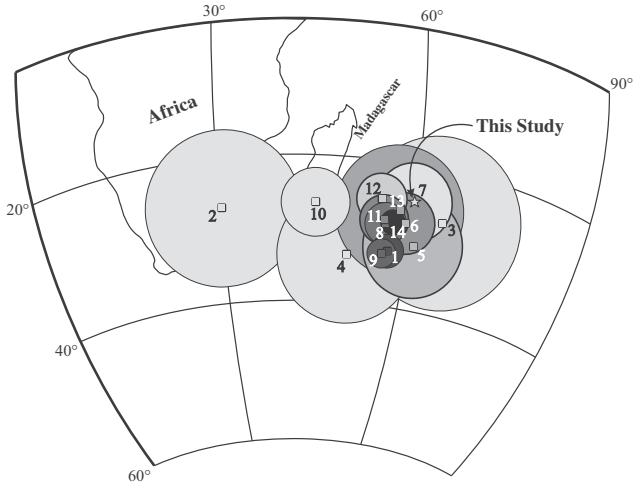


Fig. 11. African Upper Carboniferous paleomagnetic poles (see Table 2).

http://dx.doi.org/10.1016/j.jafrearsci.2013.12.006) including our new datum have been used to update our APWP (Derder et al., 2006) for Gondwana on North West African coordinates for the Paleozoic time (Table 3 as supplementary material in http://dx.doi.org/10.1016/j.jafrearsci.2013.12.006). The statistical approach (Le Goff, 1990; Le Goff et al., 1992) remains the same. It uses all data having at least part of their uncertainty windows in

Table 4
Paleozoic Gondwana APWP in NW Africa coordinates using the data as listed in Table 3 as supplementary material in http://dx.doi.org/10.1016/j.jafrearsci.2013.12.006. The results of the moscovian period are in bold.

Mean age (Ma)	*N*	λ (°S)	ϕ (°E)	A_{95}(°)	K
250	22	49.8	59.9	2.1	20
260	45	46.6	58.2	1.8	17
270	48	43.3	59.0	1.6	15
280	52	40.5	57.6	1.5	15
290	54	36.1	55.7	1.5	14
300	60	33.1	54.3	1.5	13
310	**59**	**29.4**	**51.5**	**1.8**	**11**
320	42	25.8	49.5	2.1	10
330	32	22.2	48.0	2.8	9
340	19	14.5	44.1	4.6	8
350	13	9.0	28.9	5.2	10
360	16	13.7	16.5	4.8	11
370	14	18.6	7.7	4.9	11
380	15	20.3	5.6	5.3	9
390	15	30.3	4.1	7.1	6
400	10	43.4	3.8	9.2	4
410	11	26.4	−6.5	7.7	5
420	9	23.9	−7.3	7.9	5
430	13	14.7	−8.9	7.6	7
440	8	6.1	−8.1	6.3	10
450	8	−32.5	−1.0	5.7	10
460	14	−33.0	1.0	3.5	13
470	17	−31.3	2.4	3.0	13
480	23	−29.7	2.0	2.6	14
490	36	−25.2	2.1	2.5	13
500	43	−19.1	-0.5	2.7	9

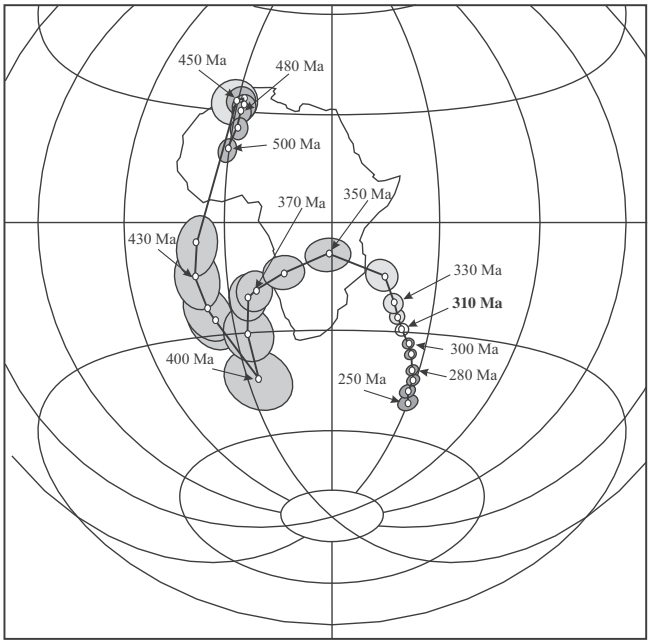


Fig. 12. APWP for Gondwana on North West African coordinates for the Paleozoic time, see Table 4.



Fig. 13. Paleogeographic reconstruction for the Moscovian (310 Ma) using the data in North West African coordinates of Laurussia (Domeier et al., 2012) and Gondwana (this study) to restore the landmasses independently.

times within a window of 20 My around the considered age. Each data is weighted by the part of its uncertainty window included in the considered 20 My window (Le Goff et al., 2002). Only a secondary weighting by the number of studied sites for each pole was added to improve the curve, as in Derder et al. (2006). The pole at 310 Ma obtained from the updated APWP corresponds to the Mean Paleomagnetic pole of Gondwana (λ = S, ϕ = 51.5°E, K = 11, A_{95} = 1.8°; – Table 4 – Fig. 12) for the Moscovian times is very close to the Africa one. It could be thus a reliable reference datum to analyze the continental drift.

7. Conclusion

Pangea reconstruction is still subject of debates (for review, see Domeier et al., 2012; Stampfli et al., 2013). For the Moscovian times, our new Gondwana pole, combined with the Laurussia one (λ = 32.5°S, ϕ = 46.6°E, A_{95} = 6.3°) translated in North West African coordinates from Domeier et al. (2012) yields a more constrained reconstruction (Fig. 13). Though the previous reconstruction was based only on the West Gondwana (Domeier et al., 2012), the uncertainty on the location of the Gondwana pole becomes more reduced, A_{95} decreasing from 3.3° (4.3° for the inclination-corrected data assumed as more realistic in the cited paper) to 1.8°. Improvement of this reconstruction needs now new paleomagnetic works or dating improvement in Laurussia. There are in fact few well-dated formations of this age in Laurussia because most of deposits correspond to continental facies. However, better-specified paleontological or radiometric age determinations as well as new paleomagnetic studies in the few well-dated marine formations in this super-continent are the key to have a reliable reconstruction for this key period of the Hercynian times.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jafrearsci.2013.12.006.

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